

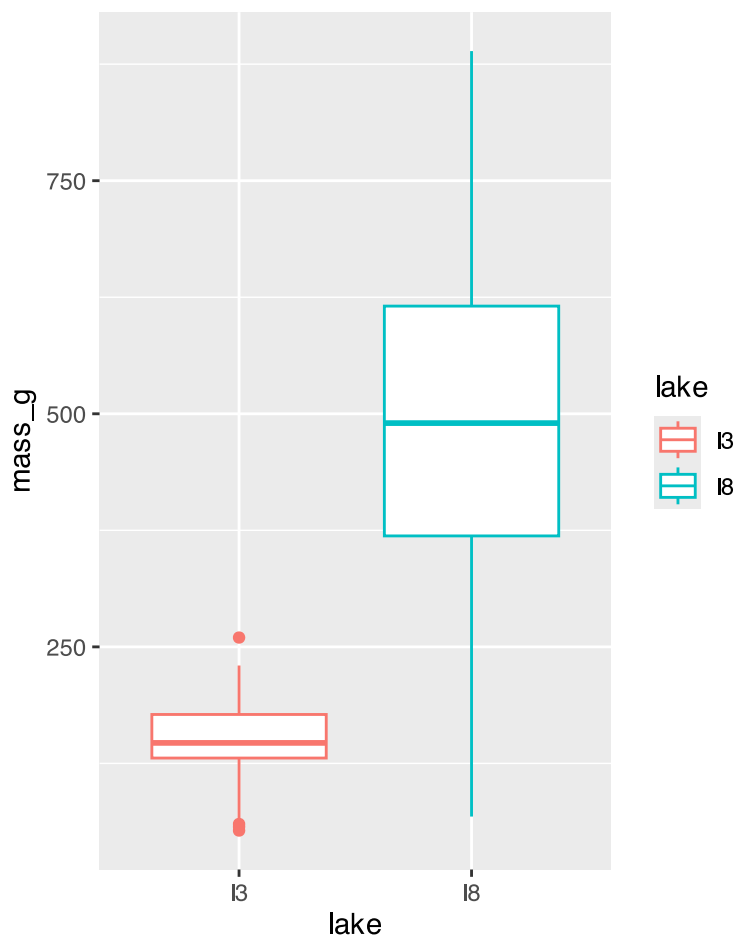
# Lecture 09 Correlation and Regression

Bill Perry

## Lecture 8: Review

Covered

- Study design
- Causality in ecology
- Experimental design:
  - Replication, controls, randomization, independence
- Sampling in field studies
- Power analysis: *a priori* and *post hoc*
- Study design and analysis



## Lecture 9: Overview

### The objectives:

This lecture covers two fundamental statistical techniques in biology: correlation and regression analysis. Based on Chapters 16-17 from Whitlock & Schluter's *The Analysis of Biological Data* (3rd edition), we'll explore:

- Correlation analysis: measuring relationships between variables

- The distinction between correlation and regression
- Simple linear regression: predicting one variable from another
- Estimating and interpreting regression parameters
- Testing assumptions and handling violations
- Analysis of variance in regression
- Model selection and comparison

## Lecture 9: Correlation vs. Regression:

### What's the Difference?

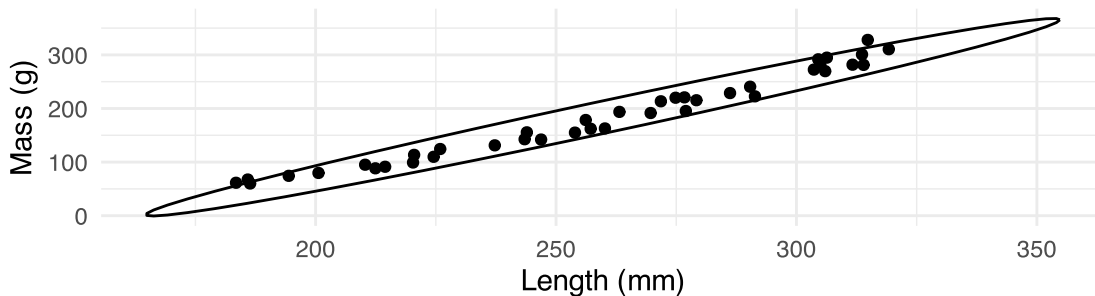
#### Correlation Analysis:

- Measures the strength and direction of a relationship between two numerical variables
- Both X and Y are random variables (both measured, neither manipulated)
- Variables are typically on equal footing (either could be X or Y)
- No cause-effect relationship implied
- Quantifies the degree to which variables are related
- Expressed as a correlation coefficient ( $r$ ) from  $-1$  to  $+1$

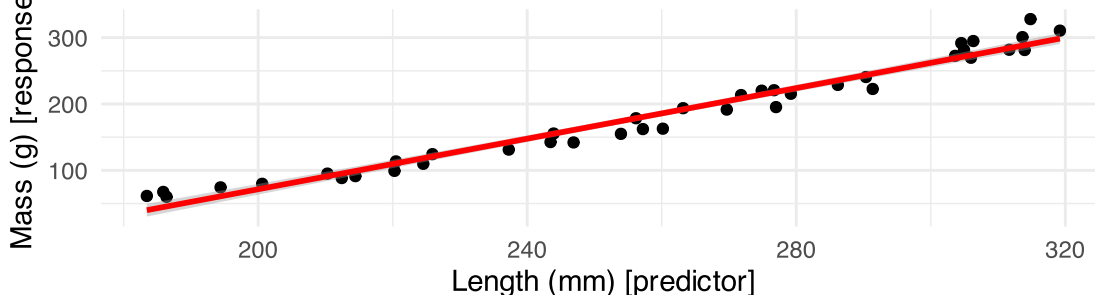
#### Regression Analysis:

- Predicts one variable (Y) from another (X)
- X is often fixed or controlled (manipulated)
- Y is the response variable of interest
- Often implies a cause-effect relationship
- Produces an equation for prediction
- Estimates slope and intercept parameters

Correlation View - no dependent/independent distinction



Regression View - predict mass from length, clear  $X \rightarrow Y$  relations



## Lecture 9: Correlation Analysis

### What Is Correlation?

**Correlation analysis** measures the strength and direction of a relationship between two numerical variables:

- Ranges from -1 to +1
- +1 indicates perfect positive correlation
- 0 indicates no correlation
- -1 indicates perfect negative correlation

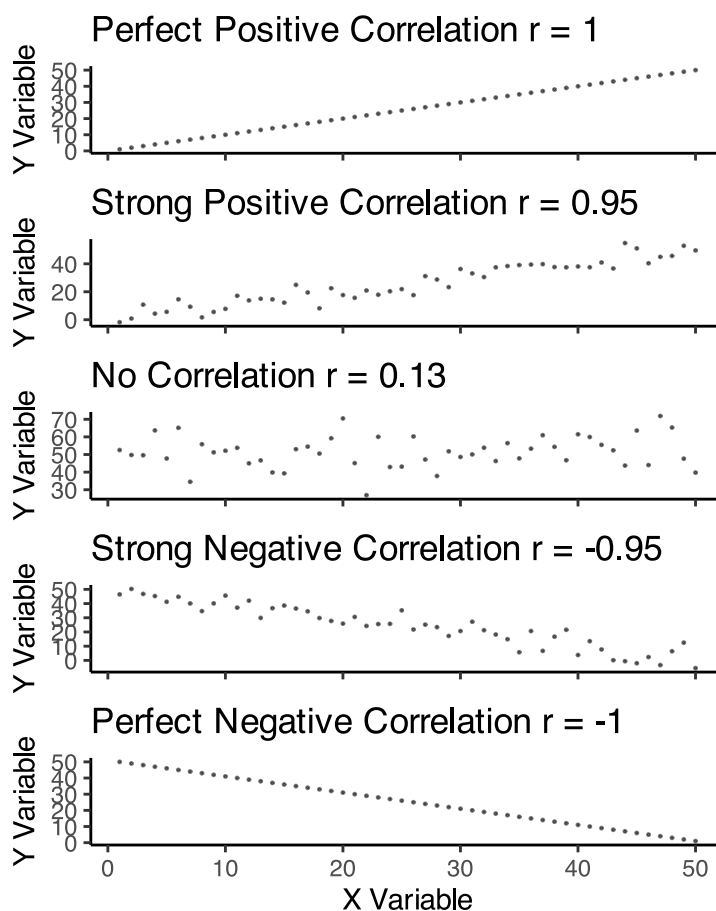
The **Pearson correlation coefficient (r)** is defined as:

$$r = \frac{\sum_i (X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum_i (X_i - \bar{X})^2 \sum_i (Y_i - \bar{Y})^2}}$$

This can be simplified as:

$$r = \frac{\text{Covariance}(X, Y)}{s_X \cdot s_Y}$$

Where  $s_X$  and  $s_Y$  are the standard deviations of X and Y.



## Lecture 9: Correlation Analysis

### Example 16.1: Flipping the Bird

Nazca boobies (*Sula granti*) - Do aggressive behaviors as a chick predict future aggressive behavior as an adult?

- correlation is  $r = 0.534$  - moderate positive relationship
- p-value = 0.007 correlation is statistically significant.

For a Pearson correlation coefficient (r) of 0.53372:

- This is r (not rho as Spearman nonparticipant below), as indicated by “cor” in your output
- To determine the amount of variation explained, you square this value:  $r^2 = 0.53372^2 = 0.2849$  (or approximately 28.49%)
- means about 28.49% of the variance in one variable can be explained by the other variable

**Note**  $t = \frac{r}{SE_r}$

```
[1] 0.5337225
```

Pearson's product-moment correlation

```
data: booby_data$visits_as_nestling and booby_data$future_aggression
t = 2.9603, df = 22, p-value = 0.007229
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 0.1660840 0.7710999
sample estimates:
      cor
0.5337225
```

## Lecture 9: Correlation Analysis

### Example 16.1: Flipping the Bird

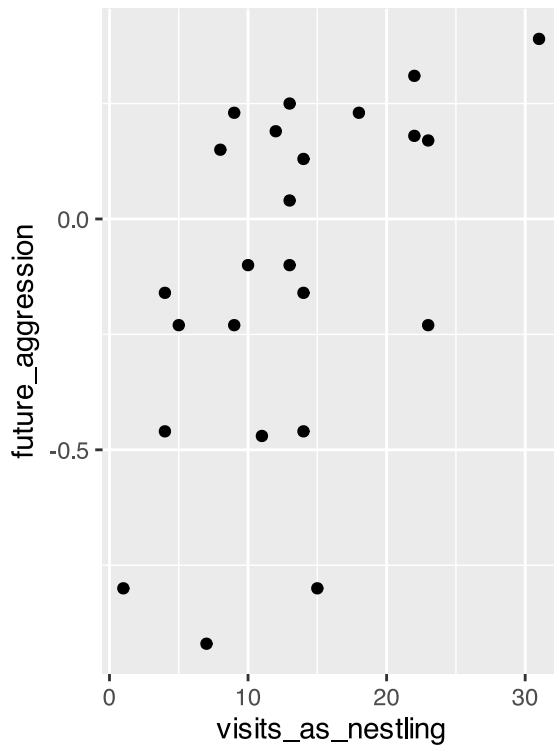
**Interpretation:** The correlation coefficient of  $r = 0.534$  suggests that Nazca boobies who experienced more visits from non-parent adults as nestlings tend to display more aggressive behavior as adults. This supports the hypothesis that early experiences influence adult behavior patterns in this species.

#### Standard Error:

$$SE_r = \sqrt{\frac{1-r^2}{n-2}}$$

**SE = 0.180**

Need to be sure relationship is not curved - note below



## Lecture 9: Correlation Analysis

### Testing Assumptions for Correlation

As described in Section 16.3, correlation analysis has key assumptions:

1. **Random sampling:** Observations should be a random sample from the population
2. **Bivariate normality:** Both variables follow a normal distribution, and their joint distribution is bivariate normal
3. **Linear relationship:** The relationship between variables is linear, not curved

Let's check these assumptions using the lion data from Example 17.1 Lion Noses:

Shapiro-Wilk normality test

```
data: lion_data$proportion_black
W = 0.88895, p-value = 0.003279
```

Shapiro-Wilk normality test

```
data: lion_data$age_years
W = 0.87615, p-value = 0.001615
```

## Lecture 9: Correlation Analysis

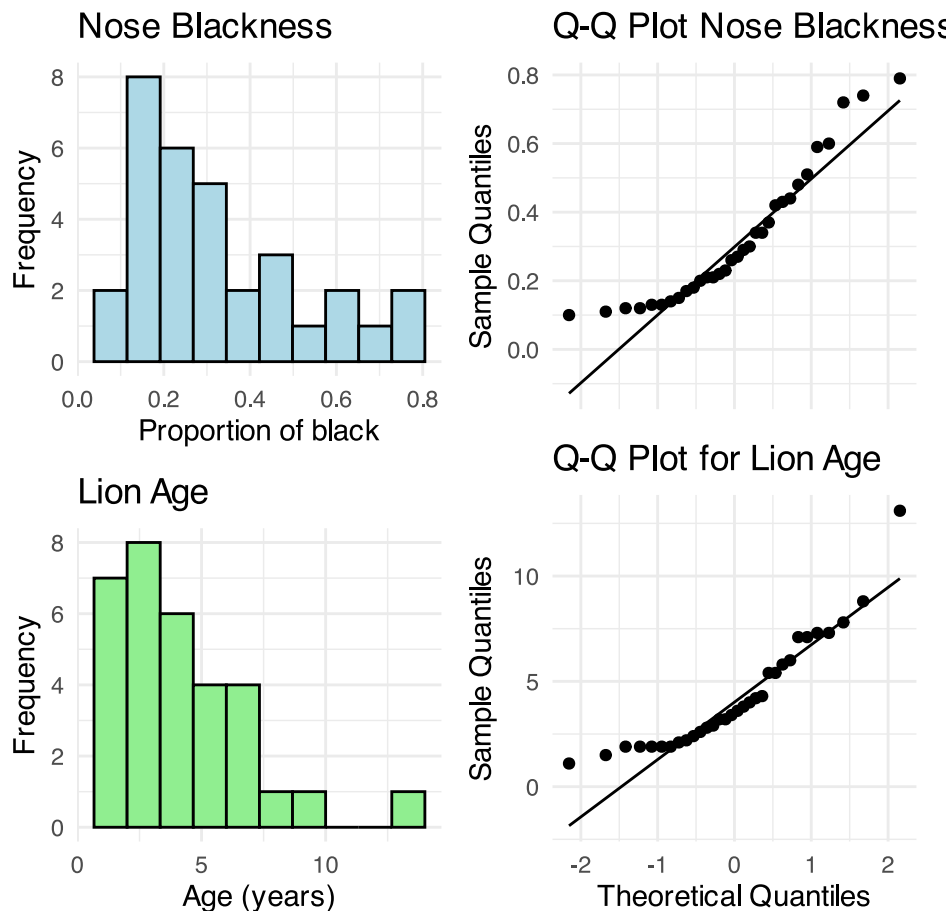
### Testing Assumptions for Correlation

As described in Section 16.3, correlation analysis has key assumptions:

1. **Random sampling:** Observations should be a random sample from the population
2. **Bivariate normality:** Both variables follow a normal distribution, and their joint distribution is bivariate normal

3. **Linear relationship:** The relationship between variables is linear, not curved

Let's check these assumptions using the lion data from Example 17.1 Lion Noses:



## Lecture 9: Correlation Analysis

**What to do if assumptions are violated:**

Transform one or both variables (log, square root, etc.)

Use non-parametric correlation (**Spearman's rank correlation**) or Kendall's tau [?](#)

Examine the data for outliers or influential points

To understand the amount of variation explained, you can square the Spearman's rho value.

For your value of 0.74485:

$$\rho^2 = 0.74485^2 = 0.5548$$

This means approximately 55.48% of the variance in ranks of one variable can be explained by the ranks of the other variable. This is similar to how  $R^2$  works in linear regression, but specifically for ranked data.

Spearman's rank correlation rho

```
data: lion_data$proportion_black and lion_data$age_years
S = 1392.1, p-value = 1.013e-06
alternative hypothesis: true rho is not equal to 0
sample estimates:
```

rho  
0.7448561

## Lecture 9: Correlation Analysis

### Correlation: Important Considerations

#### The correlation coefficient depends on the range

- Restricting range of values can reduce the correlation coefficient
- Comparing correlations between studies requires similar ranges of values

#### Measurement error affects correlation

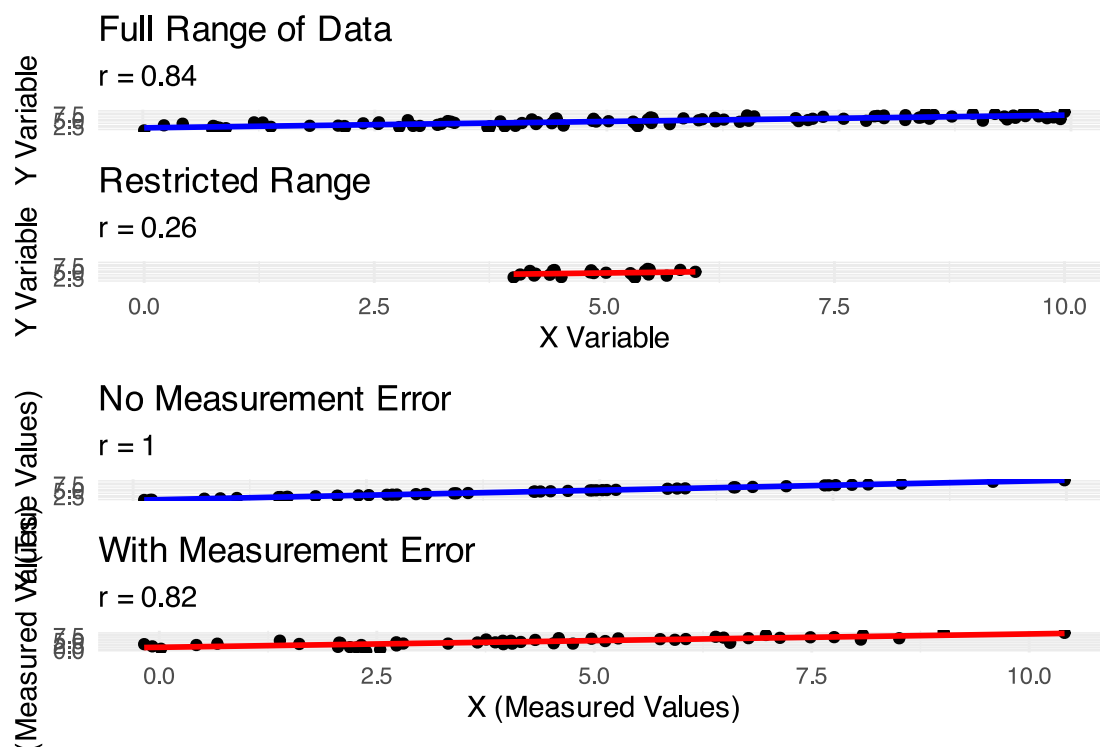
- Measurement error in X or Y tends to weaken observed correlation
- This bias is called **attenuation**
- True correlation typically stronger than observed correlation

#### Correlation vs. Causation

- Correlation does not imply causation
- Three possible explanations for correlation:
  1. X causes Y
  2. Y causes X
  3. Z (a third variable) causes both X and Y

#### Correlation significance test

- $H_0: \rho = 0$  (no correlation in population)
- $H_1: \rho \neq 0$  (correlation exists in population)
- **Test statistic:**  $t = r / SE(r)$  with  $df = n-2$



## Lecture 9: Linear Regression

## Simple Linear Regression Model

**Simple linear regression** models the relationship between a response variable (Y) and a predictor variable (X).

The **population** regression model

$$Y = \alpha + \beta X + \varepsilon$$

Where:

- Y is the response variable
- X is the predictor variable
- $\alpha$  (alpha) is the intercept (value of Y when X=0)
- $\beta$  (beta) is the slope (change in Y per unit change in X)
- $\varepsilon$  (epsilon) is the error term (random deviation from the line)

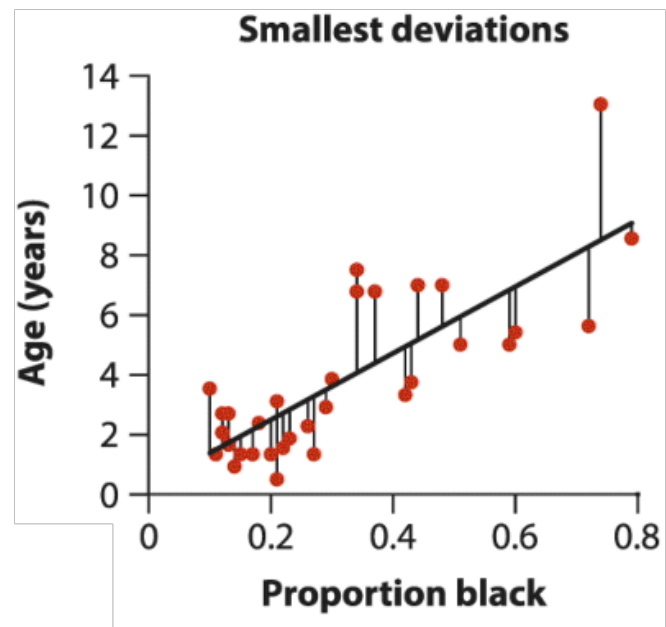
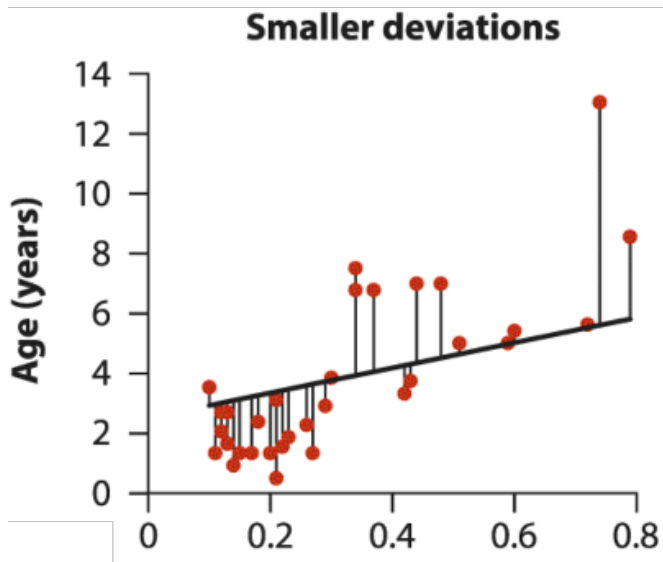
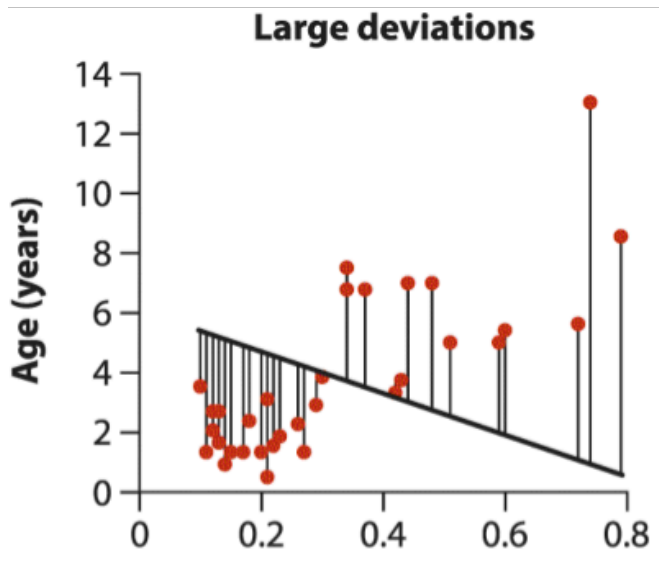
The **sample** regression equation is:

$$\hat{Y} = a + bX$$

Where:

- $\hat{Y}$  is the predicted value of Y
- a is the estimate of  $\alpha$  (intercept)
- b is the estimate of  $\beta$  (slope)

**Method of Least Squares:** The line is chosen to minimize the sum of squared vertical distances (residuals) between observed and predicted Y values.



## Lecture 9: Linear Regression

## Simple Linear Regression Model

**Simple linear regression** models the relationship between a response variable (Y) and a predictor variable (X).

The **population** regression model is:

$$Y = \alpha + \beta X + \varepsilon$$

Where:

- Y is the response variable
- X is the predictor variable
- $\alpha$  (alpha) is the intercept (value of Y when X=0)
- $\beta$  (beta) is the slope (change in Y per unit change in X)
- $\varepsilon$  (epsilon) is the error term (random deviation from the line)

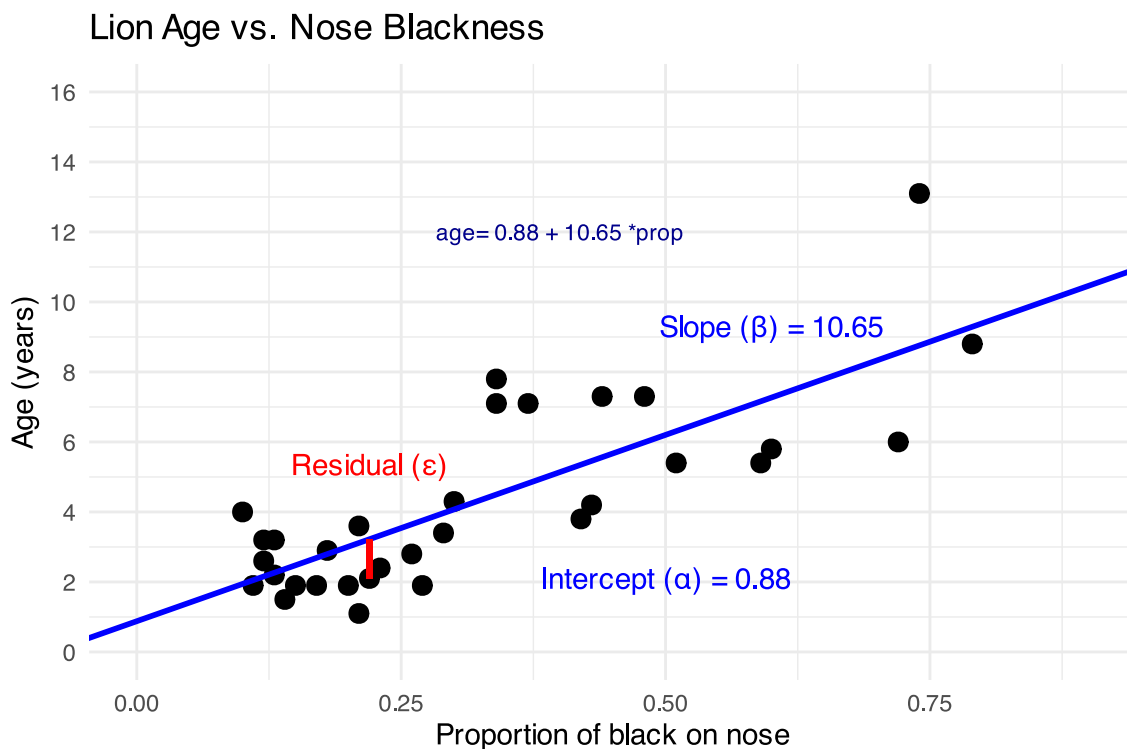
The **sample** regression equation is:

$$\hat{Y} = a + bX$$

Where:

- $\hat{Y}$  is the predicted value of Y
- a is the estimate of  $\alpha$  (intercept)
- b is the estimate of  $\beta$  (slope)

**Method of Least Squares:** The line is chosen to minimize the sum of squared vertical distances (residuals) between observed and predicted Y values.

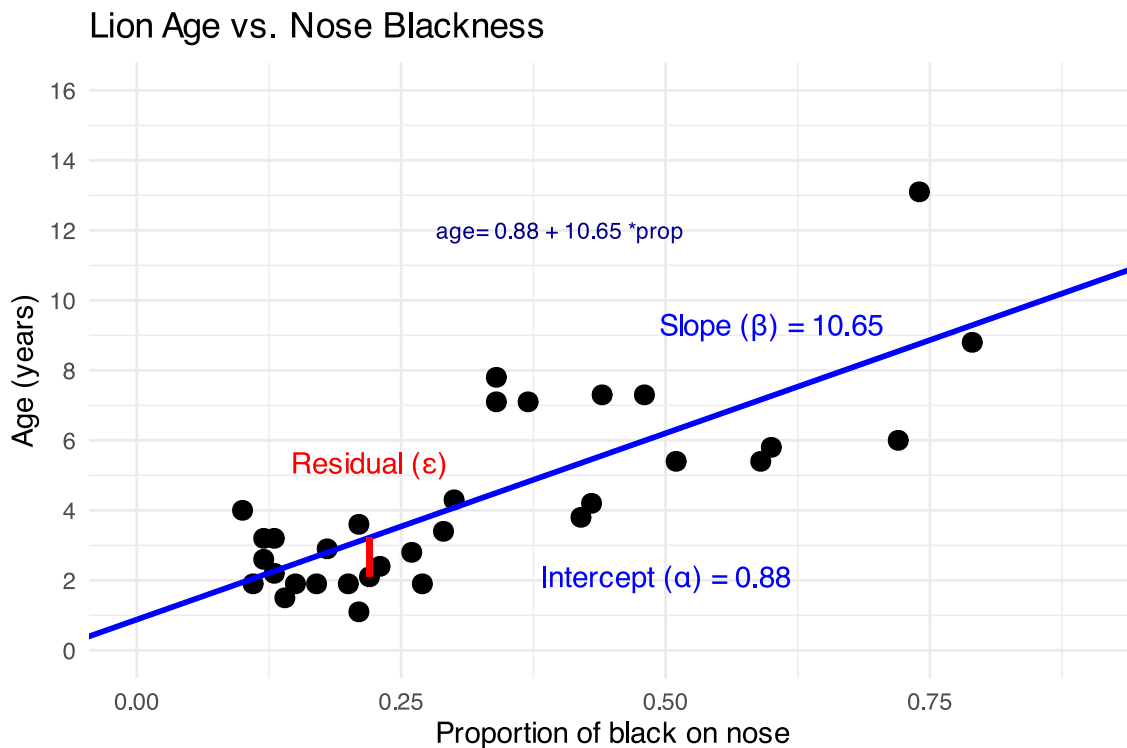


## Lecture 9: Linear Regression

From Example 17.1 in the textbook the regression line for the lion data is:

$$\text{age} = 0.88 + 10.65 \times \text{proportion}_{\text{black}}$$

This means: - When a lion has no black on its nose (proportion = 0), its predicted age is 0.88 years - For each 0.1 increase in the proportion of black, age increases by 1.065 years - The slope (10.65) indicates that lions with more black on their noses tend to be older



## Lecture 9: Linear Regression

### Simple Linear Regression Model

- male lions develop more black pigmentation on their noses as they age.
- can be used to estimate the age of lions in the field.

Call:

```
lm(formula = age_years ~ proportion_black, data = lion_data)
```

Residuals:

| Min     | 1Q      | Median  | 3Q     | Max    |
|---------|---------|---------|--------|--------|
| -2.5449 | -1.1117 | -0.5285 | 0.9635 | 4.3421 |

Coefficients:

|                  | Estimate | Std. Error | t value | Pr(> t )     |
|------------------|----------|------------|---------|--------------|
| (Intercept)      | 0.8790   | 0.5688     | 1.545   | 0.133        |
| proportion_black | 10.6471  | 1.5095     | 7.053   | 7.68e-08 *** |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.669 on 30 degrees of freedom

Multiple R-squared: 0.6238, Adjusted R-squared: 0.6113

F-statistic: 49.75 on 1 and 30 DF, p-value: 7.677e-08

## Lecture 9: Linear Regression

## Simple Linear Regression Model

The calculation for slope (b) is:

$$b = \frac{\sum_i (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_i (X_i - \bar{X})^2}$$

Given:  $\bar{X} = 0.3222$   $\bar{Y} = 4.3094$   $\sum_i (X_i - \bar{X})^2 = 1.2221$   $\sum_i (X_i - \bar{X})(Y_i - \bar{Y}) = 13.0123$

$b = 13.0123 / 1.2221 = 10.647$

Intercept (a):  $a = \bar{Y} - b\bar{X} = 4.3094 - 10.647(0.3222) = 0.879$

### Making predictions:

To predict the age of a lion with 0.50 proportion of black on its nose:

$$\hat{Y} = 0.88 + 10.65(0.50) = 6.2 \text{ years}$$

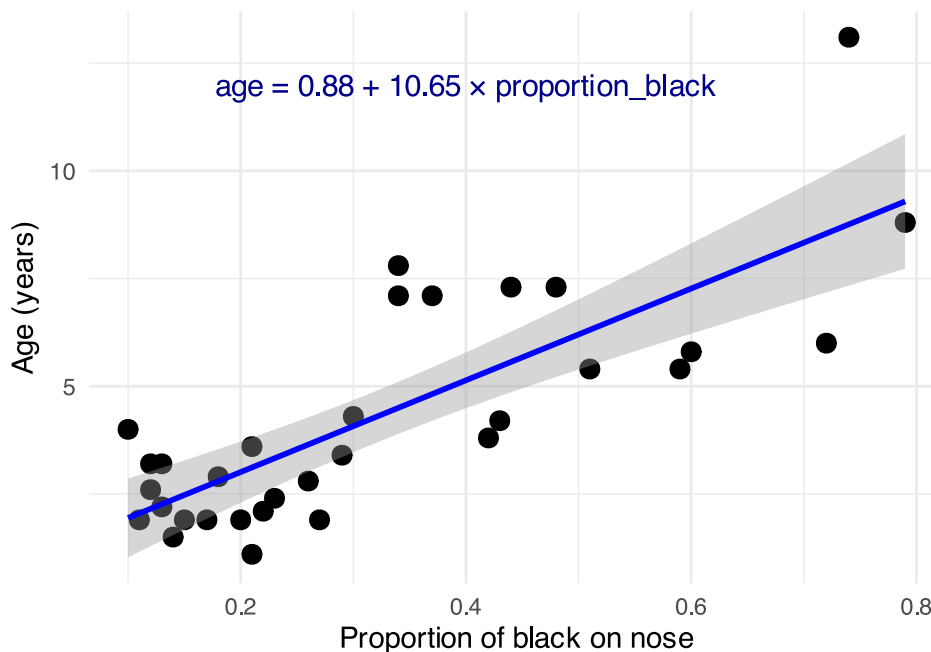
### Confidence intervals vs. Prediction intervals:

- **Confidence interval:** Range for the mean age of all lions with 0.50 black
- **Prediction interval:** Range for an individual lion with 0.50 black

Both intervals are narrowest near  $\bar{X}$  and widen as X moves away from the mean.

### Lion Age vs. Nose Blackness

Using nose pigmentation to estimate age



## Lecture 9: Linear Regression

### Example Prairie Home Companion

- Does biodiversity affect ecosystem stability?
- Tilman et al. (2006) investigated using experimental plots varying plant species

```
# A tibble: 6 × 2
  species_number log_stability
```

|   | <dbl> | <dbl> |
|---|-------|-------|
| 1 | 1     | 0.763 |
| 2 | 1     | 1.45  |
| 3 | 1     | 1.51  |
| 4 | 1     | 0.747 |
| 5 | 1     | 0.983 |
| 6 | 1     | 1.12  |

Call:

```
lm(formula = log_stability ~ species_number, data = prairie_data)
```

Residuals:

| Min      | 1Q       | Median   | 3Q      | Max     |
|----------|----------|----------|---------|---------|
| -0.82774 | -0.25344 | -0.00426 | 0.27498 | 0.75240 |

Coefficients:

|                | Estimate | Std. Error | t value | Pr(> t )     |
|----------------|----------|------------|---------|--------------|
| (Intercept)    | 1.252629 | 0.041023   | 30.535  | < 2e-16 ***  |
| species_number | 0.025984 | 0.004926   | 5.275   | 4.28e-07 *** |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.3433 on 159 degrees of freedom

Multiple R-squared: 0.149, Adjusted R-squared: 0.1436

F-statistic: 27.83 on 1 and 159 DF, p-value: 4.276e-07

```
[1] "rsquared is: 0.148953385305455"
```

Analysis of Variance Table

Response: log\_stability

|                | Df  | Sum Sq  | Mean Sq | F value | Pr(>F)        |
|----------------|-----|---------|---------|---------|---------------|
| species_number | 1   | 3.2792  | 3.2792  | 27.829  | 4.276e-07 *** |
| Residuals      | 159 | 18.7358 | 0.1178  |         |               |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

## Lecture 9: Linear Regression

The hypothesis test asks whether the slope equals zero:

- $H_0: \beta = 0$  (species number does not affect stability)
- $H_1: \beta \neq 0$  (species number does affect stability)

**The test statistic is:**  $t = \frac{b - \beta_0}{SE_b}$

With  $df = n - 2 = 161 - 2 = 159$

**Interpretation:**

The slope estimate is 0.033, indicating that log stability increases by 0.033 units for each additional plant species in the plot.

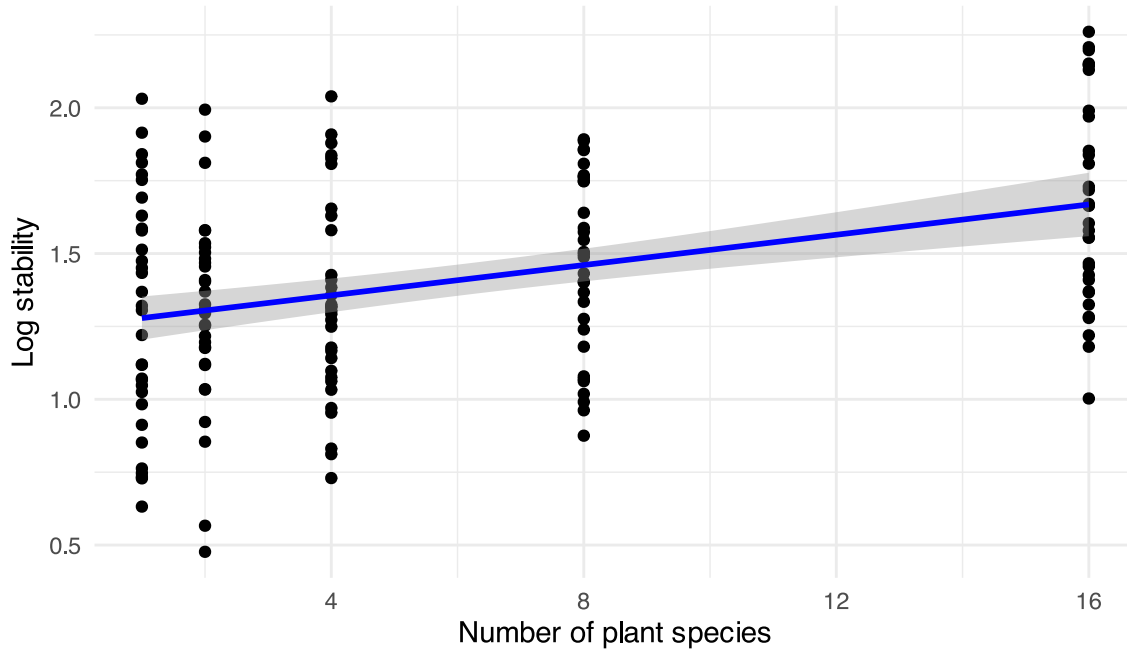
The p-value is very small ( $2.73e-10$ ), providing strong evidence to reject the null hypothesis that species number has no effect on ecosystem stability.

$R^2 = 0.222$ , meaning that approximately 22.2% of the variation in log stability is explained by the number of plant species.

This supports the biodiversity-stability hypothesis: more diverse plant communities have more stable biomass production over time.

### Biodiversity and Ecosystem Stability

$R^2 = 0.149$



## Lecture 9: Linear Regression

### Testing Regression Assumptions

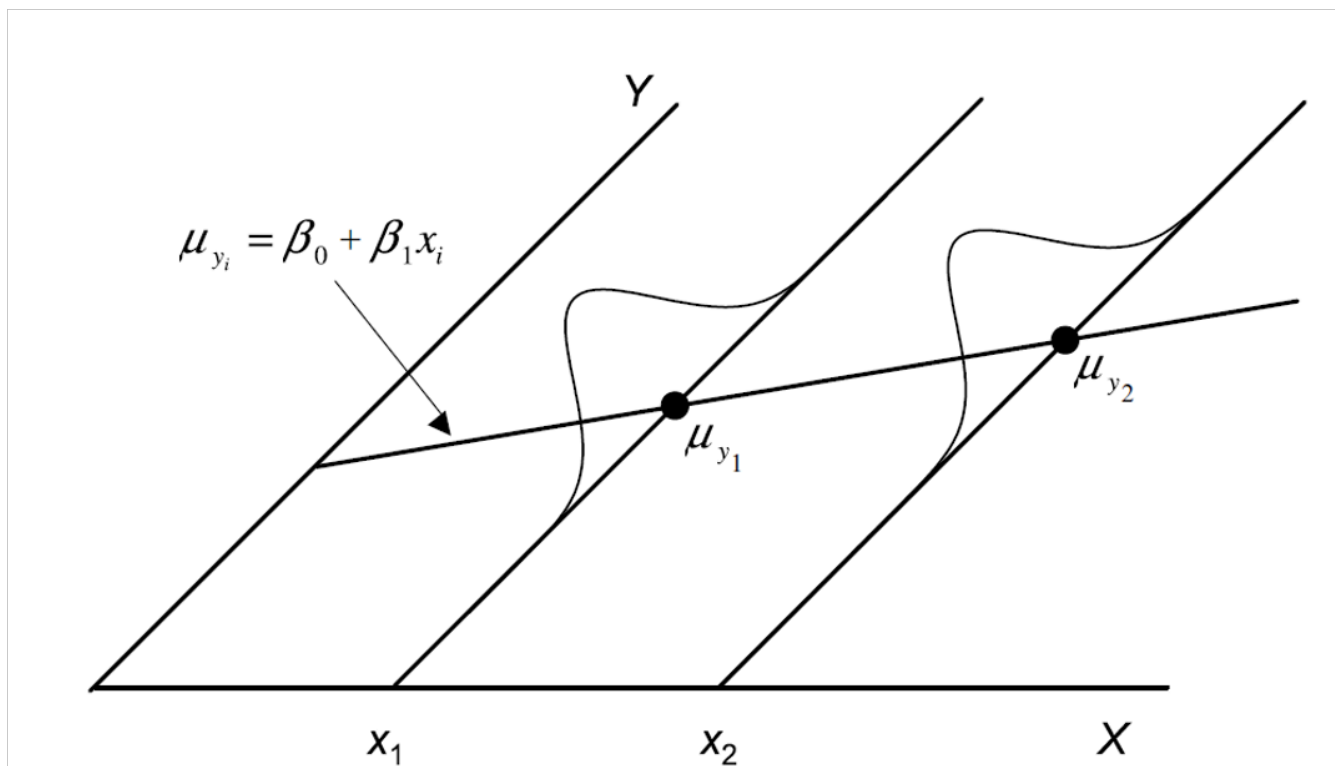
linear regression has four key assumptions:

1. **Linearity:** The relationship between X and Y is linear
2. **Independence:** Observations are independent
3. **Homoscedasticity:** Equal variance across all values of X
4. **Normality:** Residuals are normally distributed

Let's check these assumptions for the lion regression model:

Assume that **error**  $e_i$  is  $e_i = y_i - \hat{y}_i$

- normally distributed for each  $x_i$
- has the same variance
- has a mean of 0 at each  $x_i$



## Lecture 9: Linear Regression

### Testing Regression Assumptions

linear regression has four key assumptions:

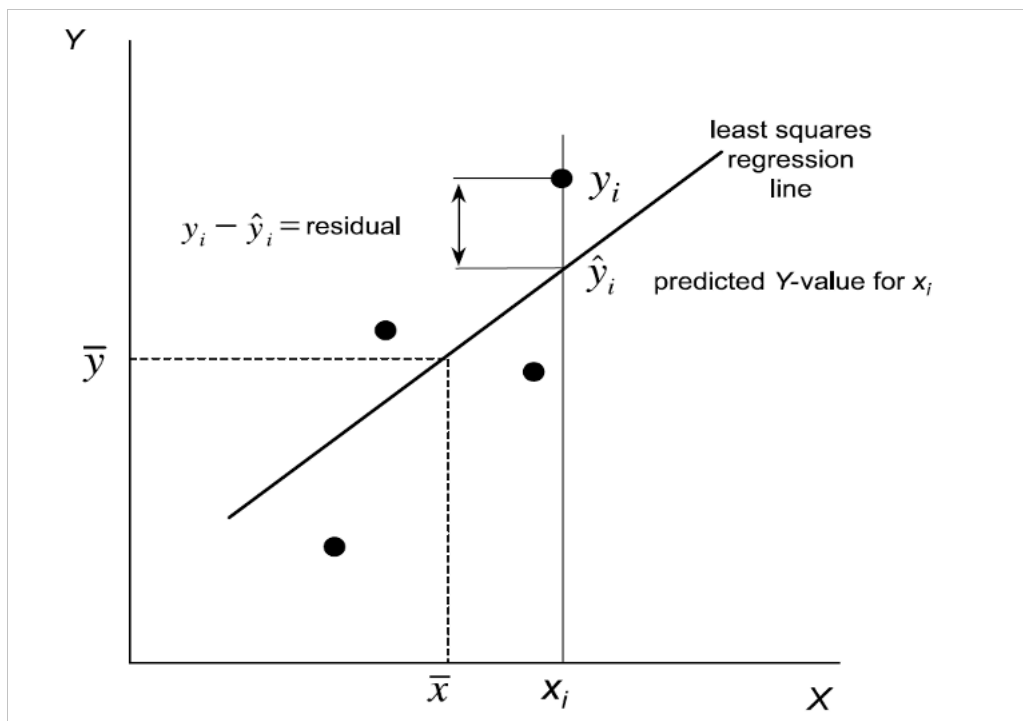
1. **Linearity:** The relationship between X and Y is linear
2. **Independence:** Observations are independent
3. **Homoscedasticity:** Equal variance across all values of X
4. **Normality:** Residuals are normally distributed

Let's check these assumptions for the lion regression model:

Assume that **error**  $\epsilon_i$  is - estimated as the residuals:  $e_i = y_i - \hat{y}_i$

- ordinary least square estimates a and b or slope and intercept to minimize the sum of the residuals squared or Mean Squared Error (MSE) as

$$\sum_{i=1}^n (y_i - \hat{y}_i)^2$$



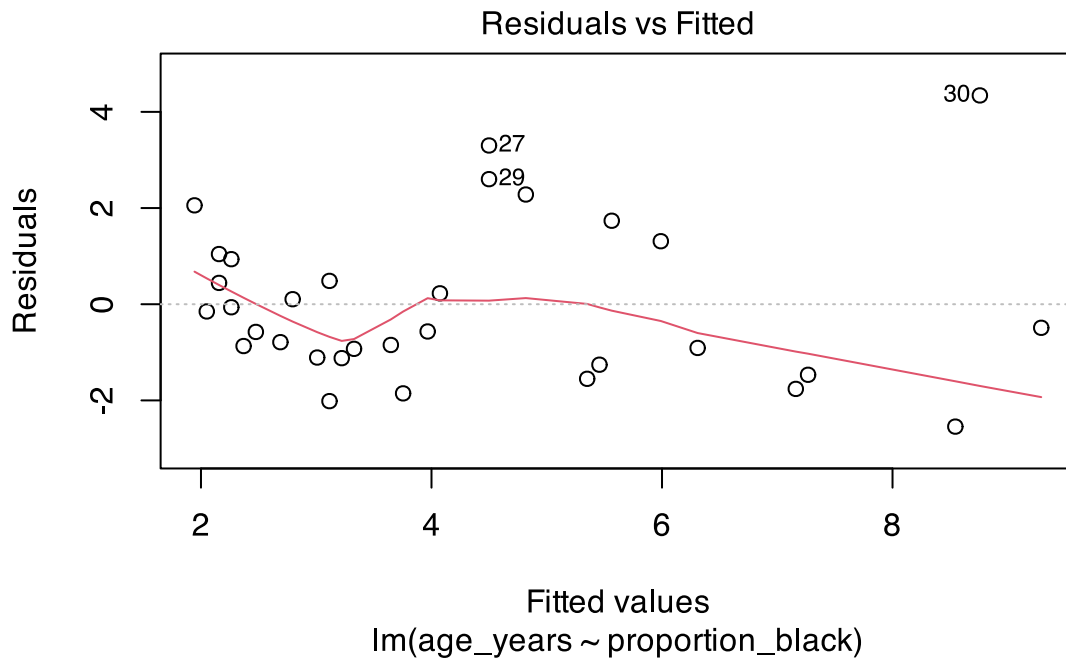
## Lecture 9: Linear Regression

### Testing Regression Assumptions

linear regression has four key assumptions:

1. **Linearity:** The relationship between X and Y is linear
2. **Independence:** Observations are independent
3. **Homoscedasticity:** Equal variance across all values of X
4. **Normality:** Residuals are normally distributed

Let's check these assumptions for the lion regression model:



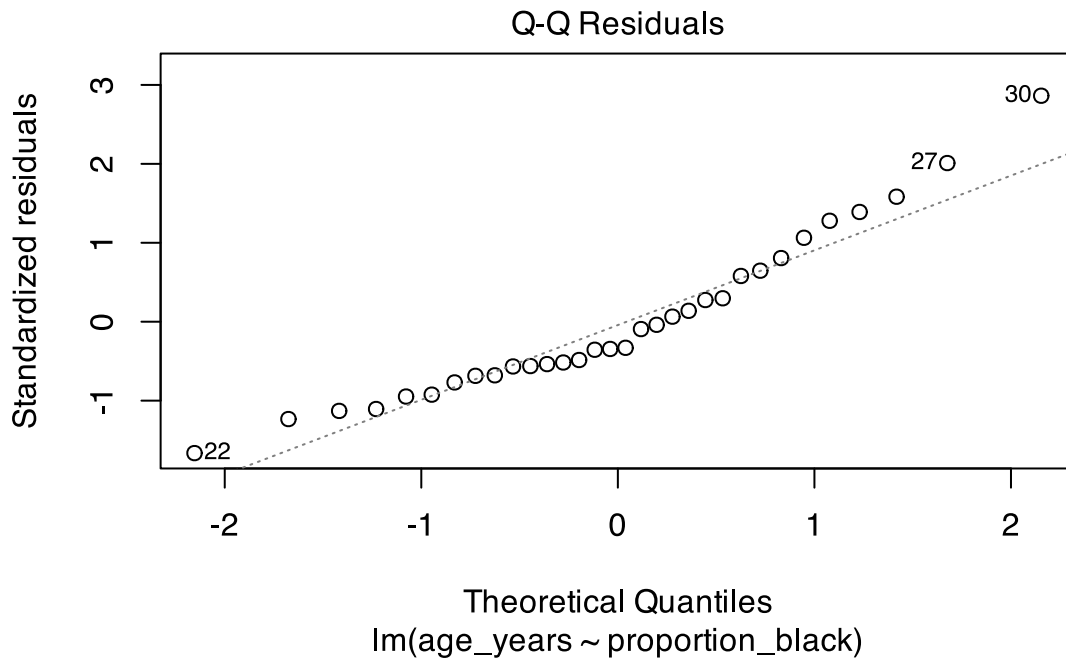
## Lecture 9: Linear Regression

### Testing Regression Assumptions

linear regression has four key assumptions:

1. **Linearity:** The relationship between X and Y is linear
2. **Independence:** Observations are independent
3. **Homoscedasticity:** Equal variance across all values of X
4. **Normality:** Residuals are normally distributed

Let's check these assumptions for the lion regression model:



## Lecture 9: Linear Regression

### Testing Regression Assumptions

linear regression has four key assumptions:

1. **Linearity:** The relationship between X and Y is linear
2. **Independence:** Observations are independent
3. **Homoscedasticity:** Equal variance across all values of X
4. **Normality:** Residuals are normally distributed

Let's check these assumptions for the lion regression model:

Shapiro-Wilk normality test

```
data: residuals(lion_model)
W = 0.93879, p-value = 0.0692
```

## Lecture 9: Linear Regression

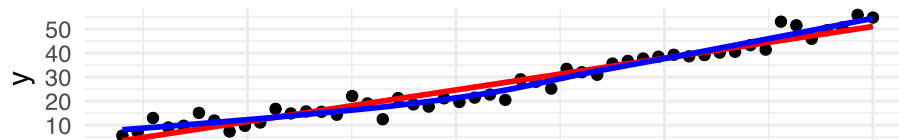
### Simple Linear Regression Model

linear regression has four key assumptions:

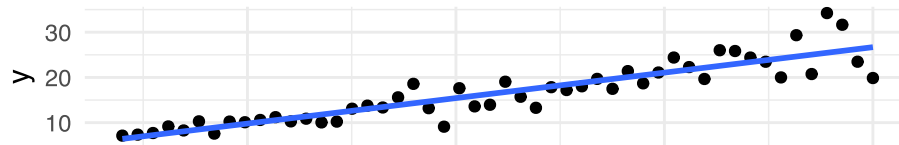
1. **Linearity:** The relationship between X and Y is linear
2. **Independence:** Observations are independent
3. **Homoscedasticity:** Equal variance across all values of X
4. **Normality:** Residuals are normally distributed

If assumptions are violated: 1. Transform the data (Section 17.6) 2. Use weighted least squares for heteroscedasticity 3. Consider non-linear models (Section 17.8)

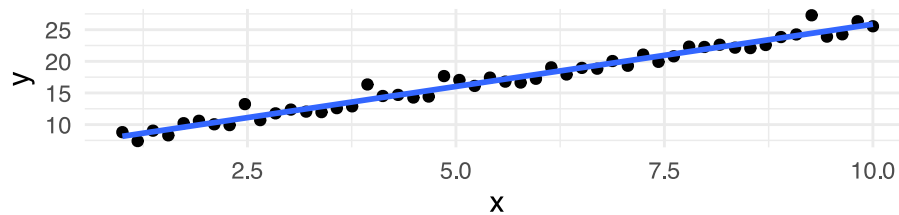
Violation: Non-linearity , Red = linear model / Blue = true



Violation: Heteroscedasticity Variance increases with x



Violation: Non-normal residuals - Residuals have skewness



## Lecture 9: Linear Regression - estimates of error and significance

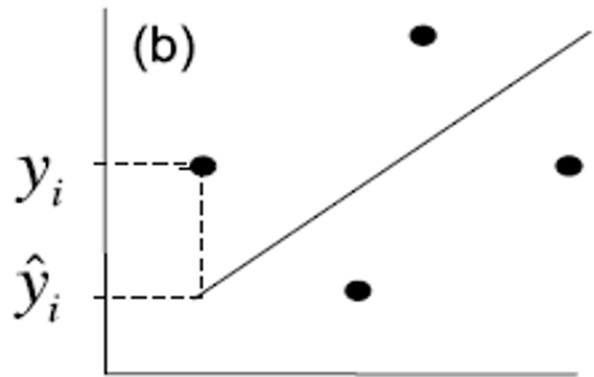
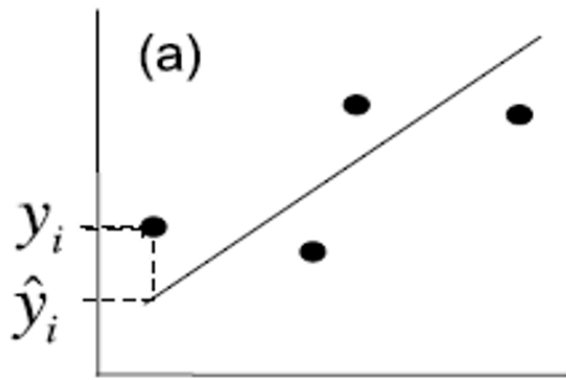
- Estimates of standard error and confidence intervals for slope and intercept to determine confidence bands
- the 95% confidence band will contain the true population line 95/100 under repeated sampling
- this is usually done in R

| Parameter       | OLS estimate                                                                                 | Standard error                                                                                                        |
|-----------------|----------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| $\beta_1$       | $b_1 = \frac{\sum_{i=1}^n [(x_i - \bar{x})(y_i - \bar{y})]}{\sum_{i=1}^n (x_i - \bar{x})^2}$ | $s_{b_1} = \sqrt{\frac{MS_{\text{Residual}}}{\sum_{i=1}^n (x_i - \bar{x})^2}}$                                        |
| $\beta_0$       | $b_0 = \bar{y} - b_1 \bar{x}$                                                                | $s_{b_0} = \sqrt{MS_{\text{Residual}} \left[ \frac{1}{n} + \frac{\bar{x}^2}{\sum_{i=1}^n (x_i - \bar{x})^2} \right]}$ |
| $\varepsilon_i$ | $e_i = y_i - \hat{y}_i$                                                                      | $\sqrt{MS_{\text{Residual}}} \text{ (approx.)}$                                                                       |

## Lecture 9: Linear Regression - estimates of error and significance

In addition to getting estimates of population parameters ( $\beta_0$ ,  $\beta_1$ ), want to test hypotheses about them

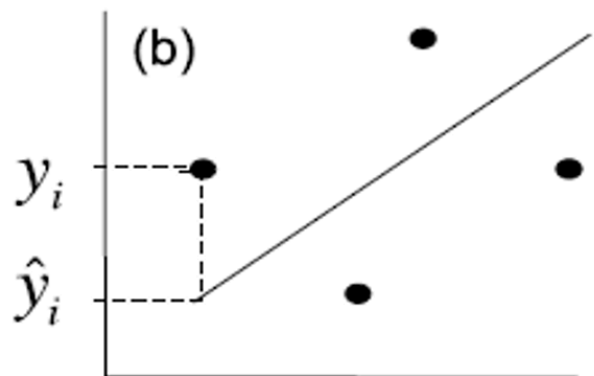
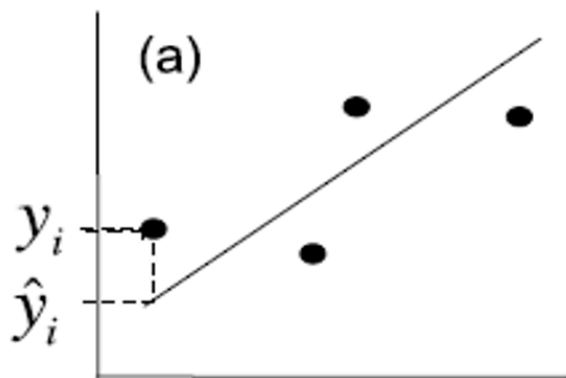
- This is accomplished by analysis of variance
- Partition variance in Y: due to variation in X, due to other things (error)



## Lecture 9: Linear Regression - estimates of variance

Total variation in Y is “partitioned” into 3 components:

- $SS_{regression}$ : variation explained by regression
  - difference between predicted values ( $\hat{y}_i$ ) and mean  $y$  ( $\bar{y}$ )
  - $dfs = 1$  for simple linear (parameters-1)
- $SS_{residual}$ : variation not explained by regression
  - difference between observed ( $y_i$ ) and predicted ( $\hat{y}_i$ ) values
  - $dfs = n - 2$
- $SS_{total}$ : total variation
  - sum of squared deviations of each observation ( $y_i$ ) from mean ( $\bar{y}$ )
  - $dfs = n - 1$



## Lecture 9: Linear Regression - estimates of variance

Total variation in Y is “partitioned” into 3 components:

- $SS_{regression}$ : variation explained by regression
  - difference between predicted values ( $\hat{y}_i$ ) and mean  $y$  ( $\bar{y}$ )
  - $dfs = 1$  for simple linear (parameters-1)
- $SS_{residual}$ : variation not explained by regression
  - difference between observed ( $y_i$ ) and predicted ( $\hat{y}_i$ ) values
  - $dfs = n - 2$
- $SS_{total}$ : total variation
  - sum of squared deviations of each observation ( $y_i$ ) from mean ( $\bar{y}$ )

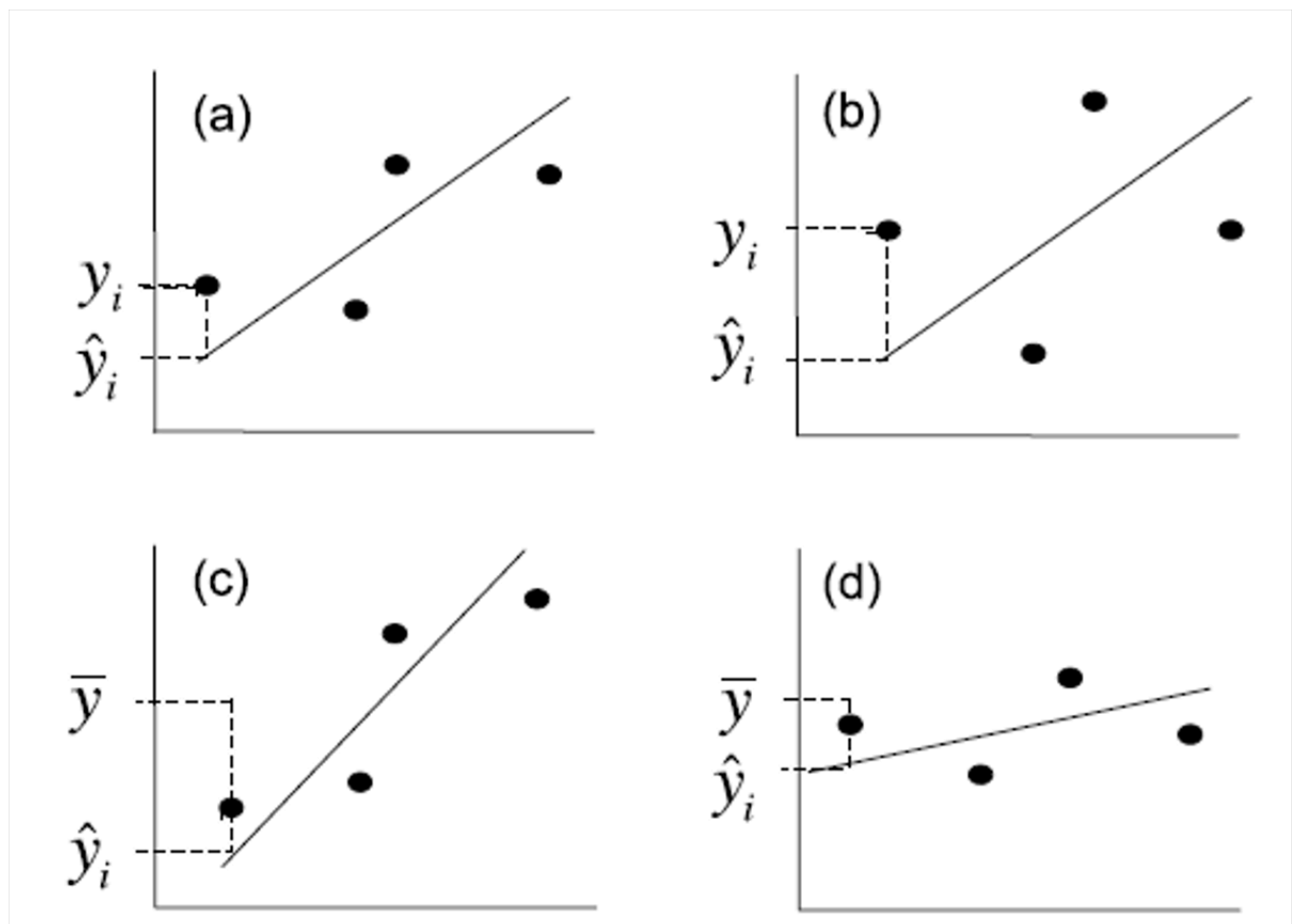
▸  $dfs = n-1$

| Source of variation | SS                                     | df    | MS                                               | Expected mean square                                              |
|---------------------|----------------------------------------|-------|--------------------------------------------------|-------------------------------------------------------------------|
| Regression          | $\sum_{i=1}^n (\hat{y}_i - \bar{y})^2$ | 1     | $\frac{\sum_{i=1}^n (\hat{y}_i - \bar{y})^2}{1}$ | $\sigma_\varepsilon^2 + \beta_1^2 \sum_{i=1}^n (x_i - \bar{x})^2$ |
| Residual            | $\sum_{i=1}^n (y_i - \hat{y}_i)^2$     | $n-2$ | $\frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{n-2}$   | $\sigma_\varepsilon^2$                                            |
| Total               | $\sum_{i=1}^n (y_i - \bar{y})^2$       | $n-1$ |                                                  |                                                                   |

## Lecture 9: Linear Regression - estimates of variance

Total variation in Y is “partitioned” into 3 components:

- $SS_{regression}$ : variation explained by regression
  - GREATER IN C than D
- $SS_{residual}$ : variation not explained by regression
  - GREATER IN B THAN A
- $SS_{total}$ : total variation



## Lecture 9: Linear Regression - estimates of variance

Sums of Squares and degrees of freedom are:

$$SS_{regression} + SS_{residual} = SS_{total}$$

$$df_{regression} + df_{residual} = df_{total}$$

- Sums of Squares depends on n
- We need a different estimate of variance

| Source of variation | SS                                     | df      | MS                                               | Expected mean square                                              |
|---------------------|----------------------------------------|---------|--------------------------------------------------|-------------------------------------------------------------------|
| Regression          | $\sum_{i=1}^n (\hat{y}_i - \bar{y})^2$ | 1       | $\frac{\sum_{i=1}^n (\hat{y}_i - \bar{y})^2}{1}$ | $\sigma_\varepsilon^2 + \beta_1^2 \sum_{i=1}^n (x_i - \bar{x})^2$ |
| Residual            | $\sum_{i=1}^n (y_i - \hat{y}_i)^2$     | $n - 2$ | $\frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{n - 2}$ | $\sigma_\varepsilon^2$                                            |
| Total               | $\sum_{i=1}^n (y_i - \bar{y})^2$       | $n - 1$ |                                                  |                                                                   |

## Lecture 9: Linear Regression - estimates of variance

Sums of Squares converted to Mean Squares

- Sums of Squares divided by degrees of freedom - does not depend on n
- $MS_{residual}$ : estimate population variation
- $MS_{regression}$ : estimate pop variation and variation due to X-Y relationship
- Mean Squares are not additive

| Source of variation | SS                                     | df      | MS                                               | Expected mean square                                              |
|---------------------|----------------------------------------|---------|--------------------------------------------------|-------------------------------------------------------------------|
| Regression          | $\sum_{i=1}^n (\hat{y}_i - \bar{y})^2$ | 1       | $\frac{\sum_{i=1}^n (\hat{y}_i - \bar{y})^2}{1}$ | $\sigma_\varepsilon^2 + \beta_1^2 \sum_{i=1}^n (x_i - \bar{x})^2$ |
| Residual            | $\sum_{i=1}^n (y_i - \hat{y}_i)^2$     | $n - 2$ | $\frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{n - 2}$ | $\sigma_\varepsilon^2$                                            |
| Total               | $\sum_{i=1}^n (y_i - \bar{y})^2$       | $n - 1$ |                                                  |                                                                   |